Assignment-3 MERN Lab

1.Write a React JSX program to display “Hello World!” by separating the HTML structure into index.html and the JSX code into App.jsx. Use React, ReactDOM, and Babel Standalone to execute the JSX in the browser. Also demonstrate how to run the program using npx http-server.

index.html

```
<!DOCTYPE html>
<html>
<head>
  <meta charset="utf-8">
  <title>React Hello World</title>

  <script src="https://unpkg.com/react@16/umd/react.development.js"></script>
  <script
src="https://unpkg.com/react-dom@16/umd/react-dom.development.js"></script>
  <script src="https://unpkg.com/@babel/standalone@7/babel.min.js"></script>
</head>

<body>
  <div id="contents"></div>

  <script type="text/babel" src="App.jsx"></script>
</body>
</html>
```

App.jsx

```
const element = (
  <div title="Outer div">
    <h1>Hello World!</h1>
  </div>
);

ReactDOM.render(
  element,
  document.getElementById("contents")
);
```

Output

Hello World!

2.Write a React JSX program to accept a number from the user and check whether the given number is prime or not when the user clicks the “Check” button. Use separate index.html and App.jsx files.

index.html

```
<!DOCTYPE html>
<html>
<head>
   <meta charset="utf-8">
   <title>Prime Number</title>

   <script src="https://unpkg.com/react@16/umd/react.development.js"></script>
   <script
src="https://unpkg.com/react-dom@16/umd/react-dom.development.js"></script>
   <script src="https://unpkg.com/@babel/standalone@7/babel.min.js"></script>
</head>

<body>

   <div id="contents"></div>

   <script type="text/babel" src="App.jsx"></script>

</body>
</html>
```

App.jsx

```
function PrimeChecker() {
   const [number, setNumber] = React.useState("");
   const [result, setResult] = React.useState("");

   function checkPrime() {
      let num=Number(number);
      let prime=true;

      if (num < 2) {
         prime=false;
      }
      for (let i = 2; i < num; i++) {
         if (num % i === 0) {
            prime = false;
```

```
            break;
          }
        }
        if (prime) {
          setResult(" is a prime number.");
        } else {
          setResult(" is not a prime number.");
        }
      }
      return (
        <div>
          <h1>Prime Number Checker</h1>
          <input
            type="number"
            value={number}
            onChange={(e) =>
              setNumber(e.target.value)}

          />

          <br /><br />

          <button onClick={checkPrime}>
            Check
          </button>

          <h2>{result}</h2>
        </div>
      );
}
const element=<PrimeChecker />
ReactDOM.render(
  element,
  document.getElementById("content")
);
```

Output

Enter number: 7
Prime Number

Enter number: 10
Not a Prime Number

3.Write a React JSX program to accept a number from the user and display its reverse when the user clicks the “Reverse” button. Use separate index.html and App.jsx files.

index.html

```
<!DOCTYPE html>
<html>
<head>
   <meta charset="utf-8">
   <title>Reverse Number</title>

   <script src="https://unpkg.com/react@16/umd/react.development.js"></script>
   <script
src="https://unpkg.com/react-dom@16/umd/react-dom.development.js"></script>
   <script src="https://unpkg.com/@babel/standalone@7/babel.min.js"></script>
</head>

<body>

   <div id="contents"></div>

   <script type="text/babel" src="App.jsx"></script>

</body>
</html>
```

App.jsx

```
class App extends React.Component {
   constructor() {
      super();

      this.state = {
         number: "",
         result: ""
      };
   }

   reverseNumber = () => {
      let n = this.state.number;

      let reverse = n.split("").reverse().join("");

      this.setState({
         result: reverse
      });
```

```
  };

  render() {
    return (
      <div>
        <h2>Reverse a Number</h2>

        <input
          type="number"
          value={this.state.number}
          onChange={(e) =>
            this.setState({ number: e.target.value })
          }
        />

        <button onClick={this.reverseNumber}>
          Reverse
        </button>

        <h3>Reverse: {this.state.result}</h3>
      </div>
    );
  }
}

ReactDOM.render(
  <App />,
  document.getElementById("contents")
);
```

Output

```
Enter number: 12345
Reverse: 54321
```

4.Write a React JSX program to display “Hello World!” using a separate App.jsx file stored in the src folder and an index.html file stored in the public folder. Use Babel CLI and @babel/preset-react to compile the JSX code into JavaScript. Demonstrate the complete procedure for initializing the project, installing the required Babel packages, compiling the JSX, and executing the application using npx http-server.

public/index.html

```
<!DOCTYPE html>
<html>
<head>
   <meta charset="utf-8">
   <title>React JSX Transform</title>

   <script src="https://unpkg.com/react@16/umd/react.development.js"></script>
   <script
src="https://unpkg.com/react-dom@16/umd/react-dom.development.js"></script>
</head>

<body>

   <div id="contents"></div>

   <script src="App.js"></script>

</body>
</html>
```

src/App.jsx

```
const element = (
   <div title="Outer div">
      <h1>Hello World!</h1>
   </div>
);

ReactDOM.render(
   element,
   document.getElementById("contents")
);
```

Output

Hello World!

5.Write a React JSX program to accept a number from the user and check whether the given number is a palindrome when the user clicks the Check button.

index.html

```
<!DOCTYPE html>
<html>
<head>
   <meta charset="utf-8">
   <title>Palindrome Number</title>

   <script src="https://unpkg.com/react@16/umd/react.development.js"></script>
   <script
src="https://unpkg.com/react-dom@16/umd/react-dom.development.js"></script>
   <script src="https://unpkg.com/@babel/standalone@7/babel.min.js"></script>
</head>

<body>

   <div id="contents"></div>

   <script type="text/babel" src="App.jsx"></script>

</body>
</html>
```

App.jsx

```
class App extends React.Component {
   constructor() {
      super();

      this.state = {
         number: "",
         result: ""
      };
   }

   checkPalindrome = () => {
      const number = this.state.number;

      const reverse = number.split("").reverse().join("");

      if (number === reverse) {
         this.setState({
            result: "Palindrome Number"
         });
      } else {
         this.setState({
            result: "Not a Palindrome Number"
         });
```

```
    }
  };

  render() {
    return (
      <div>
        <h2>Palindrome Checker</h2>

        <input
          type="number"
          value={this.state.number}
          onChange={(e) =>
            this.setState({ number: e.target.value })
          }
        />

        <button onClick={this.checkPalindrome}>
          Check
        </button>

        <h3>{this.state.result}</h3>
      </div>
    );
  }
}

ReactDOM.render(
  <App />,
  document.getElementById("contents")
);
```

Output

121
Palindrome Number

123
Not a Palindrome Number

6.Write a React JSX program to accept temperature in Celsius and convert it into Fahrenheit when the user clicks the Convert button.

index.html

```
<!DOCTYPE html>
```

```
<html>
<head>
  <meta charset="utf-8">
  <title>Celsius to Fahrenheit</title>

  <script src="https://unpkg.com/react@16/umd/react.development.js"></script>
  <script
src="https://unpkg.com/react-dom@16/umd/react-dom.development.js"></script>
  <script src="https://unpkg.com/@babel/standalone@7/babel.min.js"></script>
</head>

<body>

  <div id="contents"></div>

  <script type="text/babel" src="App.jsx"></script>

</body>
</html>
```

App.jsx

```
class App extends React.Component {
  constructor() {
    super();

    this.state = {
      celsius: "",
      fahrenheit: ""
    };
  }

  convert = () => {
    const c = Number(this.state.celsius);

    const f = (c * 9 / 5) + 32;

    this.setState({
      fahrenheit: f
    });
  };

  render() {
    return (
      <div>
        <h2>Celsius to Fahrenheit</h2>

        <input
          type="number"
```

```
            value={this.state.celsius}
            onChange={(e) =>
              this.setState({
                celsius: e.target.value
              })
            }
          />

          <button onClick={this.convert}>
            Convert
          </button>

          <h3>
            Fahrenheit: {this.state.fahrenheit}
          </h3>
        </div>
      );
    }
}

ReactDOM.render(
  <App />,
  document.getElementById("contents")
);
```

Output

Celsius: 25
Convert
Fahrenheit: 77